Algorithm-Driven Deep Learning Framework for Parallel MRI Reconstruction

Supplementary Material

I. List of Notation

We mainly use u to denote data in the image domain, t, v to denote the data in the framelet transform domain, and x, y, z to denote data in the K-space (Fourier) domain. The iteration result at step k in an algorithm is indicated typically as $(\cdot)^k$, e.g., $x^k, y^k, z^k, u^k, t^k, v^k$. The main notations used throughout this paper are summarized in Tab. I.

TABLE I
List of Notation

	Symbol	**Description**
Scalars	$h \times w$	MRI data dimensions per coil image
	c	Number of coils
	n	$n = h \times w \times c$, total number of K-space elements across all coil images
	N	Number of the filters in the 2-level 3DHSTF decomposition
Special Matrices	I	The identity matrix
	F	The matrix in $\mathbb{C}^{n\times n}$ w.r.t. the Fourier transform on a vector of size n
	P	The diagonal binary sampling matrix in $\mathbb{R}^{n\times n}$
	Q	Diagonal binary matrix in $\mathbb{R}^{n\times n}$, indicating the absence positions in K-space, $Q = I - P$
	Γ	Diagonal matrix in $\mathbb{R}^{Nn\times Nn}$ containing the regularization parameters on its diagonal
	x	The vectorized of full K-space data $x \in \mathbb{C}^n$ of all coils
	g	The observed undersampled and noisy K-space data, $g = Px + \epsilon \in \mathbb{C}^n$
	G	SPIRiT interpolation kernel matrix, $G \in \mathbb{R}^{n\times n}$
	W_{3D}	The transformation matrix of the 3DHSTF system, $W_{3D} \in \mathbb{R}^{Nn\times n}$ and $W_{3D}^\top W_{3D} = I$
ℓ_1-W3D-Re	z	The real-valued counterpart of g, defined as $z = \begin{bmatrix}\mathrm{Re}(g)\\ \mathrm{Im}(g)\end{bmatrix} \in \mathbb{R}^{2n}$
	$\mathcal{G}$	The matrix $\mathcal{G} \in \mathbb{R}^{2n\times 2n}$ is derived from $G - I$ as $\mathcal{G} = \begin{bmatrix}G-I & 0\\ 0 & G-I\end{bmatrix}$
	$\mathcal{Q}$	The matrix $\mathcal{Q} \in \mathbb{R}^{2n\times 2n}$ is derived from Q as $\mathcal{Q} = \begin{bmatrix}Q & 0\\ 0 & Q\end{bmatrix}$
	$\mathcal{F}^{-1}$	The matrix $\mathcal{F}^{-1} \in \mathbb{R}^{2n\times 2n}$ is derived from F^{-1} as $\mathcal{F}^{-1} = \begin{bmatrix}\mathrm{Re}(F^{-1}) & -\mathrm{Im}(F^{-1})\\ \mathrm{Im}(F^{-1}) & \mathrm{Re}(F^{-1})\end{bmatrix}$
	$\mathcal{W}$	The matrix $\mathcal{W} \in \mathbb{R}^{2Nn\times 2n}$ is derived from W_{3D} as $\mathcal{W} = \begin{bmatrix}W_{3D} & 0\\ 0 & W_{3D}\end{bmatrix}$
DSM	$\mathcal{N}_\Theta(\cdot,\cdot)$	The DSM network
	β	The iteration step size
	σ	The Gaussian noise level
	γ	A constant parameter used to ensure the decrease of β and σ
	$N(u, \sigma^2 I)$	Gaussian distribution with mean u and variance σ^2

II. SPIRiT Interpolation Kernel

In SPIRiT, the goal is to learn a convolutional operator (i.e., the SPIRiT operator) that maps a neighborhood in K-space across multiple coils to the center point of that neighborhood in each coil. The ACS serves as training data for learning the SPIRiT operator.

Suppose the number of the coils is c. The 3D convolution kernel tensor across coils and space, denoted by $\mathcal{K}$, has dimensions $s \times s \times c \times c$. In particular, $\mathcal{K}_{:,:,:,\ell}$ represents the

3D interpolation kernel used to reconstruct the ℓ-th coil from all coils. We estimate this convolution kernel from the fully sampled autocalibration signal (ACS) data, see [1], [2].

After estimating $\mathcal{K}$, we embed it into a larger tensor of size $h \times w \times c \times c$ by padding with zeros. This padding reflects the fact that certain spatial offsets are not utilized. For notational convenience, we continue to denote the padded kernel as $\mathcal{K}$.

We now describe the construction of the matrix G in SPIRiT model using the tensor $\mathcal{K}$. Let $X \in \mathbb{C}^{h\times w\times c}$ be a tensor that collects all coil images, i.e., $X_{:,:,\ell}$ is the ℓ-th coil image x_ℓ. The tensor Y, which has the same dimensions as X, is obtained by convolving X with the 3D convolution kernel tensor $\mathcal{K}$, given by

$$Y_{i,j,\ell} = \sum_{k=1}^{c}\sum_{p=1}^{h}\sum_{q=1}^{w} X_{i+p,j+q,k}\mathcal{K}_{p,q,k,\ell}.$$

This operation can be written in matrix-vector form as:

$$\mathrm{vec}(Y) = G\mathrm{vec}(X),$$

where $\mathrm{vec}(X) \in \mathbb{C}^{hwc}$ is the flattened column vector version of X, with elements ordered such that $X_{i,j,k}$ is the $((k-1)hw + (j-1)h + i)$-th entry of $\mathrm{vec}(X)$, $\mathrm{vec}(Y)$ is defined in the same way, G is the global convolution matrix, defined block-wise as:

$$G = \begin{bmatrix} G_{11} & G_{12} & \cdots & G_{1c} \\ G_{21} & G_{22} & \cdots & G_{2c} \\ \vdots & \vdots & \ddots & \vdots \\ G_{c1} & G_{c2} & \cdots & G_{cc} \end{bmatrix}.$$

Here, each block G_{pq} is a doubly circulant matrix representing convolution with the 2D kernel slice $\mathcal{K}_{:,:,p,q}$. Specifically, for a matrix $K \in \mathbb{C}^{h\times w}$, the associated doubly circulant matrix is constructed as

$$\begin{bmatrix} \mathrm{circ}(K_{1,:)} & \mathrm{circ}(K_{2,:)} & \cdots & \mathrm{circ}(K_{h,:)} \\ \mathrm{circ}(K_{h,:)} & \mathrm{circ}(K_{1,:)} & \cdots & \mathrm{circ}(K_{h-1,:)} \\ \vdots & \vdots & \ddots & \vdots \\ \mathrm{circ}(K_{2,:)} & \mathrm{circ}(K_{3,:)} & \cdots & \mathrm{circ}(K_{1,:)} \end{bmatrix},$$

where $\mathrm{circ}(a)$ is the circulant matrix with its first row as a.

With this construction, we are now in a position to analyze the singular values of G. For each pair $(p,q) \in [h] \times [w]$, define the matrix $P^{(p,q)} \in \mathbb{C}^{c\times c}$ by

$$P_{ij}^{(p,q)} = (F_0^\top \mathcal{K}_{:,:,i,j} F_0)_{pq},$$

where F is the discrete Fourier transform matrix and $[N] := \{1, 2, \ldots, N\}$ for a positive integer N. According to [3, Theorem 6] the singular values of G are given by

$$\sigma(G) = \bigcup_{(p,q)\in[h]\times[w]} \sigma\left(P^{(p,q)}\right),$$

and hence,

$$\|G\|_2 = \max\{\gamma : \gamma \in \sigma(G)\}.$$

For further details, we refer the reader to [3].

III. Directional 3D Haar Semi-Tight Framelet System

We introduce the 3D directional Haar semi-tight framelet (3DHSTF) system for W_{3D} in ℓ_1-W3D model. Note that coil images from the same imaging slice exhibit strong structural similarity. To exploit this redundancy, the 2D coil images are stacked into a 3D volume data $\boldsymbol{x}$, allowing the 3DHSTF system [1], [4], with its two-level decomposition, to effectively extract sparse features from the combined data.

Let $\boldsymbol{\delta}_k(i) := 1$ if $i = k$ and $\boldsymbol{\delta}_k(i) = 0$ if $i \neq k$ for $i = (i_1, i_2, i_3), k = (k_1, k_2, k_3) \in \mathbb{Z}^3$. That is, $\boldsymbol{\delta}_k$ is the Kronecker delta sequence with respect to the position k. The 3DHSTF system is determined by the following filters

$$\text{3DHSTF} := \{a^H; b_x, b_y, b_{xy}, b_{x,y}, b_{\text{aux}}\} \tag{1}$$

of finitely supported sequences, where a^H represents the 3D Haar low-pass filter:

$$a^H = \frac{1}{8}\left(\boldsymbol{\delta}_{(0,0,0)} + \boldsymbol{\delta}_{(0,0,1)} + \boldsymbol{\delta}_{(0,1,0)} + \boldsymbol{\delta}_{(1,0,0)} + \boldsymbol{\delta}_{(0,1,1)} + \boldsymbol{\delta}_{(1,0,1)} + \boldsymbol{\delta}_{(1,1,0)} + \boldsymbol{\delta}_{(1,1,1)}\right) \tag{2}$$

$b_x, b_y, b_{xy}, b_{x,y}$ represent high-pass filters in different directions:

$$\begin{aligned} b_x &= \frac{1}{4}\left(\boldsymbol{\delta}_{(1,0,0)} - \boldsymbol{\delta}_{(0,0,0)}\right), \quad b_{xy} = \frac{\sqrt{2}}{8}\left(\boldsymbol{\delta}_{(1,1,0)} - \boldsymbol{\delta}_{(0,0,0)}\right), \\ b_y &= \frac{1}{4}\left(\boldsymbol{\delta}_{(0,1,0)} - \boldsymbol{\delta}_{(0,0,0)}\right), \quad b_{x,y} = \frac{\sqrt{2}}{8}\left(\boldsymbol{\delta}_{(1,0,0)} - \boldsymbol{\delta}_{(0,1,0)}\right), \end{aligned} \tag{3}$$

and an auxiliary filter b_{aux} to satisfy the partition of unity condition of the tight framelet system [1]:

$$\widehat{b_{\text{aux}}}(\xi) = 1 - \left(|\widehat{a^H}(\xi)|^2 + |\widehat{b_x}(\xi)|^2 + |\widehat{b_y}(\xi)|^2 + |\widehat{b_{x,y}}(\xi)|^2 + |\widehat{b_{xy}}(\xi)|^2\right), \ \xi \in \mathbb{R}^3.$$

Here

$$\hat{\eta}(\xi) := \sum_{k \in \mathbb{Z}^3} \eta(k) e^{-\mathrm{i}k\cdot\xi}, \ \xi \in \mathbb{R}^3$$

is the Fourier series of a finitely supported filter $\eta = \{\eta(k)\}_{k\in\mathbb{Z}^3}$.

Given a multi-coil image $\boldsymbol{u} \in \mathbb{R}^{h\times w\times c}$, the $W_{3D}\boldsymbol{u}$ is given by a 2-level 3DHSTF decomposition acting on the mult-coil image $\boldsymbol{u}$:

$$\begin{aligned} W_{3D}\boldsymbol{u} =& \{\eta \circledast \boldsymbol{u} : \eta \in \{b_x, b_y, b_{x,y}, b_{xy}\}\} \\ & \cup \{(\eta \uparrow 2) \circledast (a^H \circledast \boldsymbol{u}) : \eta \in \{a^H; b_x, b_y, b_{x,y}, b_{xy}\}\}, \end{aligned}$$

where $\circledast$ denotes the circular convolution, which produces one low-pass framelet coefficient data:

$$(a^H \uparrow 2) \circledast (a^H \circledast \boldsymbol{u}) \in \mathbb{R}^{h\times w\times c}$$

and 8 high-pass framelet coefficient data:

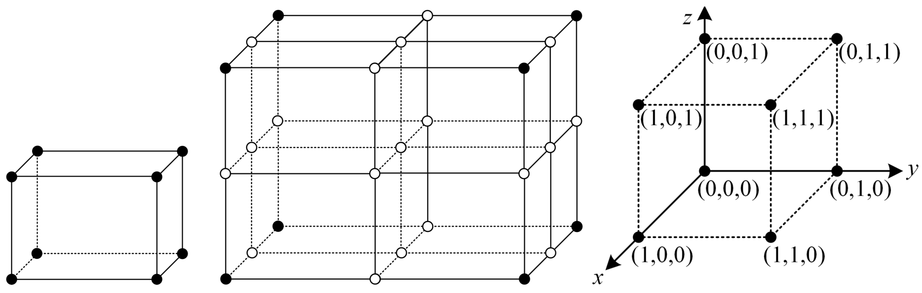
$$\left\{\eta \circledast \boldsymbol{u}, (\eta \uparrow 2) \circledast (a^H \circledast \boldsymbol{u}) : \eta \in \{b_x, b_y, b_{x,y}, b_{xy}\}\right\} \subset \mathbb{R}^{h\times w\times c}.$$

Here $\eta \uparrow 2$ is the upsampling operator (by 2) acting on a sequence. The multi-coil framelet coefficent data $W_{3D}\boldsymbol{u}$ can be viewed as a collection of framelet coefficient coil images:

$$W_{3D}\boldsymbol{u} = \{v^{\ell}_{\eta_i} \in \mathbb{R}^{h\times w} : i = 1, \ldots, 9; \ell = 1, \ldots, c\}. \tag{4}$$

Note that the auxiliary filter b_{aux} does not involve in the decomposition and regularization process as it only serves the purpose of perfect reconstruction in the reconstruction phase.

In terms of visualization, as shown in Fig. 1, the 3DHSTF decomposition considers 8 data points convolution at each level, where each black dot represents a filter value. If the current decomposition level is $j = 1, 2, \ldots$, then the spacing between data point convolution is given by $2^{j-1}-1$. In the first-layer decomposition (Fig. 1(a)), the spacing is 0, while in the second layer (Fig. 1(b)), the spacing increases to 1. This increasing in receptive field in the second layer enables 3DHSTF to capture broader contextual information, thereby enhancing feature extraction. Fig. 1(c) illustrates the coordinates of the 8 data points where 3DHSTF is applied, denoted as $\boldsymbol{\delta}_{(x,y,z)}$. The decomposition process ensures that the combined filters cover all the frequency bands and preserve the signal's structure while maintaining the perfect reconstruction property of the framelets. For more about framelets, we refer to [5]–[7].



(a) First-layer (b) Second-layer (c) 3D data directions

Fig. 1. Illustrations of first-layer and second-layer decomposition, along with 3DHSTF decomposition directions, where the xy-direction represents the coil images and the z-direction represents the stacking direction of the coil images.

IV. Proof of Theorem 1

Proof of Theorem 1. Given that the gradient of the function p is $\nabla p(y) = \mathcal{G}^\top \mathcal{G} y$, its Lipschitz constant is $L = \|\mathcal{G}\|_2^2$, where $\mathcal{G}$ is

$$\mathcal{G} = \begin{bmatrix} (G-I) & 0 \\ 0 & (G-I) \end{bmatrix}.$$

Since $\|\mathcal{G}\|_2^2 = \|\mathcal{G}^\top \mathcal{G}\|_2$, and

$$\mathcal{G}^\top \mathcal{G} = \begin{bmatrix} (G-I)^\top (G-I) & 0 \\ 0 & (G-I)^\top (G-I) \end{bmatrix}.$$

According to the properties of the matrix 2-norm, $\|\mathcal{G}^\top \mathcal{G}\|_2 = \|(G-I)^\top (G-I)\|_2 \leqslant (\|G\|_2 + 1)^2$. Therefore, the Lipschitz constant can be expressed as $L = \|\mathcal{G}^\top \mathcal{G}\|_2 \leqslant (\|G\|_2 + 1)^2$. Note that the matrix 2-norm of the SPIRiT interpolation kernel G is discussed in Sec. II. $\square$

V. Proof of Theorem 2

The class of all lower semicontinuous convex functions $f : \mathbb{R}^d \to (-\infty, +\infty]$ such that $\operatorname{dom} f := \{x \in \mathbb{C}^d : f(x) < +\infty\} \neq \emptyset$ is denoted by $\Gamma_0(\mathbb{R}^d)$. Consider the following optimization problem:

$$\arg\min_{x \in \mathbb{C}^n} p(x) + q(x) + r(Ax) \tag{5}$$

where A is a $d \times n$ matrix, $p \in \Gamma_0(\mathbb{R}^n)$ is differentiable, $q \in \Gamma_0(\mathbb{R}^n)$, and $r \in \Gamma_0(\mathbb{C}^d)$. The primal-dual three-operator splitting (PD3O) algorithm [8] provides weak convergence guarantees and allows for a larger step size, enabling faster convergence. It can be used to solve model 5, and its iterative form is as follows:

$$x^{k+1} = \operatorname{prox}_{\rho q}(y^k), \tag{6a}$$

$$z^{k+1} = \operatorname{prox}_{\delta r^*}\left((I - \rho\delta A A^\top) z^k + \delta A(2x^{k+1} - y^k - \rho \nabla p(x^{k+1}))\right), \tag{6b}$$

$$y^{k+1} = x^{k+1} - \rho \nabla p(x^{k+1}) - \rho A^\top z^{k+1}. \tag{6c}$$

One PD3O iteration can be interpreted as an operator $\mathrm{T}_{\mathrm{PD3O}}$ such that $(y^{k+1}, z^{k+1}) = \mathrm{T}_{\mathrm{PD3O}}(y^k, z^k)$. The convergence analysis of PD3O is given in the following lemma.

Lemma 1 (Sublinear convergence rate [8])**.** *Let $p \in \Gamma_0(\mathbb{R}^n)$ and its gradient be Lipschitz continuous with constant L. Choose ρ and δ such that $\rho < 2/L$ and $B = \rho\delta^{-1}(I - \rho\delta AA^\top)$ is positive definite. Let (y^*, z^*) be any fixed point of $\mathrm{T}_{\mathrm{PD3O}}$, and $\{(y^k, z^k)\}_{k\geq 0}$ be the sequence generated by PD3O. Define $\|(y,z)\|_B := \sqrt{\|y\|^2 + \langle z, Bz\rangle}$. Then, the following statements hold.*

(i) *The sequence $\{(\|(y^k, z^k) - (y^*, z^*)\|_B)\}_{k\geqslant 0}$ is monotonically nonincreasing.*
(ii) *The sequence $\{(\|(y^{k+1}, z^{k+1}) - (y^k, z^k)\|_B)\}_{k\geqslant 0}$ is monotonically nonincreasing. Moreover, $\|(y^{k+1}, z^{k+1}) - (y^k, z^k)\|_B^2 = o\left(1/(k+1)\right)$.*

Lemma 2 (Lemma 5 in [9])**.** *Let $\mathcal{A} = \mathcal{W}\mathcal{F}^{-1}$ be given in ℓ_1-W3D-Re model. Then, for any positive numbers ρ and δ, the matrix $I - \rho\delta\mathcal{A}\mathcal{A}^\top$ is positive definite if and only if $\rho\delta < 1$.*

Proof of Theorem 2. By Lemma 2 and $\mathcal{A}$ from ℓ_1-W3D-Re model, the matrix $\rho\delta^{-1}(I - \rho\delta\mathcal{A}\mathcal{A}^\top)$ is positive definite if $\rho\delta < 1$. By Lemma 1, we know that the gradient of the function $\frac{1}{2}\|\mathcal{G}y\|_2^2$ in ℓ_1-W3D-Re model is Lipschitz continuous, and its Lipschitz constant is $L = (\|G\|_2 + 1)^2$. Therefore, when the parameters ρ and δ satisfy the conditions $\rho < 2/L$ and $\rho\delta < 1$, it follows from Lemma 1 that the PD3O algorithm's convergence conditions are clearly satisfied. The proof is complete. $\square$

References

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